

QUICK START 最短主线

```
x <- sn_read(path)
x <- sn_initialize_seurat_object(x, project = 'atlas')
x <- sn_run_cluster(x, integration_method = 'harmony')
```

```
x <- sn_run_annotation(x, group_by = 'cluster', result_id = 'labels')
sn_list_results(x)
sn_plot_result(x, analysis_type = 'de', result_id = 'markers', view = 'volcano')
```

Data & I/O 数据与读写

```
sn_list_10x_paths()
发现 10x/STARsolo/spatial 输入

sn_read() / sn_write()
qs2, h5ad, BPCells 与表格 I/O

sn_set_layer_backend()
dgCMatrix <-> BPCells layer
```

Preprocess & QC 预处理

```
sn_initialize_seurat_object()
创建中心 Seurat/Shennong object

sn_add_qc_metrics()
按 assay/layer 重算 QC

sn_filter_cells() / sn_filter_genes()
过滤细胞和基因

sn_normalize_data()
log, scran 或 SCTransform

sn_find_doublets()
scDblFinder / Scrublet

sn_remove_ambient_contamination()
decontX/decontPro/SoupX
```

关键: assay/layer 明确; 样本和受试者 provenance 不要静默合并。

Verb grammar 命名语法

```
sn_list_* / sn_get_*
列举摘要 / 读取实体

sn_run_* / sn_find_* / sn_score_*
执行工作流 / 推断 / 打分

sn_check_* / sn_validate_*
诊断环境 / 断言契约

sn_assess_* / sn_calculate_* / sn_plot_*
QC 分析 / 指标 / 图形
```

Cluster & Integrate 聚类整合

```
sn_run_cluster()
聚类、参数网格、批次整合

sn_run_multimodal()
WNN/totalVI/Coralysiss/MMoChi

sn_transfer_labels()
Seurat/Coralysiss/scANVI/scArches

sn_assess_integration()
整合质量评估
```

先用 sn_list_methods(task) 查看默认后端和当前可用性。

Annotate, DE & Pathways 注释与差异

```
sn_run_annotation()
SingleR/CellTypist/PopV/...

sn_review_annotation()
低置信度证据审查

sn_find_de()
markers/contrast/pseudobulk/bulk

sn_run_enrichment()
ORA/GSEA
```

Method controls 后端控制

```
sn_list_methods(task)
方法、默认值、runtime 与 availability

sn_get_method_status(method, task)
依赖、安装动作、输入和引用

sn_get_integration_control_template()
整合后端的完整 control surface
```

Versioned result 统一结果

```
sn_list_results(object)
查看 analysis_type / result_id

sn_get_result(object, type, result_id)
读取已存结果, 不直接摸 misc

sn_validate_result(result)
校验 schema 和 primary table

sn_store_result(object, type, result_id, result)
写入规范存储

sn_audit_results(object)
审计旧结果和非注册 artifact
```

主表位于 tables\$primary; 结果身份由 analysis_type + result_id 决定。

Object anatomy 对象结构

```
Assays / layers
RNA, ADT, spatial; counts/data/custom layers

meta.data
cell, sample, donor, condition 和 labels

reductions / graphs / images
UMAP/PCA/latent/邻接图/空间坐标

misc$shennong$results
版本化结果而不是临时 list
```

Reproducibility 复现要点

```
result_id
为每个分析结果命名, 不覆盖旧结果

assay + layer
声明表达来源, 尤其 counts 与 corrected layers

sample_by + seed
保留统计重复并固定随机性
```

Composition, DA, State & CNV 组成与状态

`sn_calculate_composition()`
计数与比例

`sn_compare_composition()`
样本作为统计重复

`sn_test_abundance()`
Propeller/Milo/scCODA/permutation

`sn_prioritize_states()`
Augur/RareQ/Scissor

`sn_run_cnv()`
inferCNVpy/CopyKAT

sample-level 比较先按 sample 聚合；cell 不是生物学重复。

Programs, GRN & Metabolism 程序调控

`sn_score_programs()` / `sn_test_programs()`
UCell/AUCell/GSVA/ssGSEA/mean

`sn_discover_programs()`
NMF/cNMF/Hotspot

`sn_run_grn()`
GENIE3/SCENIC/GRNBoost2

`sn_run_metabolism()`
gene sets/scMetabolism/scFEA/Compass

Accepted inputs 输入契约

`Seurat`
单细胞、多模态与空间主入口

`matrix` / `list` / `SummarizedExperiment`
standalone bulk 入口

`sn_analysis_result`
直接进入结果可视化

`data.frame`
兼容 composition、distribution 和 association

Communication & Dynamics 通讯与动态

`sn_run_cell_communication()`
LIANA/CellChat/NicheNet/...

`sn_run_trajectory()`
Slingshot/Monocle3/Palantir

`sn_run_velocity()`
scVelo/RegVelo

`sn_run_fate()`
CellRank

Spatial 空间组学

`sn_run_spatial()`
统一 task dispatcher

`sn_find_spatial_features()`
Moran's I/nnSVG/SPARK-X

`sn_find_spatial_domains()`
BANKSY/stLearn/BayesSpace/...

`sn_run_spatial_neighborhood()`
邻域富集与共现

`sn_run_spatial_deconvolution()`
cell2location

Bulk transcriptomics Bulk 转录组

`sn_run_bulk()`
QC/DE/pathway/network/survival dispatcher

`sn_assess_bulk_qc()`
PCA、相关性与异常值

`sn_score_bulk_pathways()`
sample-level pathway scores

`sn_run_wgcna()` / `sn_run_survival()`
网络与生存

`sn_run_clinical_association()`
临床表型关联

Canonical plotting 统一绘图

`sn_list_plot_methods(type)`
发现 accepted_input, default, required_parameters

`sn_plot_result(object, ...)`
统一 object / analysis_type / result_id / view

`sn_plot_distribution()`
violin/box/histogram/density/ridge

`sn_plot_association()`
cell/sample numeric association scatter

Seurat中有多个同类结果时显式给 result_id；只有 unique/default 才自动选择。

Common result views 常用视图

`de`
volcano, ma, effect, heatmap | default: volcano

`enrichment`
dot, bar, ridge, network, emap, gsea | default: dot

`annotation`
confidence_cluster, confidence_cell, confusion, projection | default: confidence_cluster

`cell_communication`
bubble, heatmap, network, chord, river, ligand_target, comparison | default: bubble

`trajectory`
embedding, pseudotime, lineage_probability, dynamic_heatmap, gene_trend, branch_comparison | default: embedding

`cnv`
heatmap, umap, score, sample, association | default: heatmap

`metabolism`
activity, heatmap, differential, sample | default: activity

`bulk_qc`
library_size, detected_features, mean_correlation, pca, correlation_heatmap, sample_scatter | default: library_size

Publication 出版输出

`sn_list_figure_profiles()`
screen/column/page/slide

`sn_apply_figure_profile()`
应用字体、线宽和版面约束

`sn_validate_figure()`
出版前结构化检查

`sn_export_figure()`
PDF/SVG/TIFF/PNG

`sn_export_figure_bundle()`
figure + source data + manifest